# Karl Shroff

**(919) 946-7200 | krshroff126@gmail.com | Raleigh, NC | linkedin.com/in/karl-shroff**

Biomedical Engineering student with experience in medical device research, design, and validation. Adept at translating clinical needs into engineered solutions and leading teams through technical and strategic challenges. Skilled in balancing regulatory, manufacturing, and business considerations to deliver practical and compliant designs. Proven leadership in managing large organizations and executing projects effectively under pressure.

## EDUCATION

***North Carolina State University & University of North Carolina at Chapel Hill***

**MS-Biomedical Engineering –** MedTech Innovation & Entrepreneurship **May 2027**

**B.S. Biomedical Engineering –** Specialization in Medical Microdevices & Rehabilitation Engineering **May 2026**

**Relevant Coursework:** Biomaterials, Design, Microcontrollers, Bioinstrumentation, Rehabilitation Robotics, Wearables & Biosensors

## CORE COMPETENCIES

- **Medical Device Development:** FDA 510(k) pathway, IEC 60601-1 (Medical Device Safety), ISO 13485, DFMEA, Design Verification Protocols, Regulatory Documentation, TAM/SAM/SOM Market Analysis, Clinical Needs Assessment
- **Engineering & Design:** PCB Design (Altium, KiCAD), SolidWorks, Fusion 360/E-CAD, MATLAB, FEA (Ansys), Topology Optimization, Generative Design, Additive Manufacturing, DFM/CAM, Arduino (C/C++), Signal Processing, Sensor Integration
- **Leadership & Operations:** Cross-functional team management (50-400+ members), Lean Six Sigma Principles, CAPM-aligned project management, Budget management and financial planning, Stakeholder partnerships, Process improvement (72% quality & satisfaction increase)
- **Certifications and Accreditations:** Quality Management Systems and Lean Six Sigma Principles (2026), CAPM (2026), NCSU Leadership Development Program (2025), Truist Emerging Leaders Certification (2025)

## RELEVANT WORK EXPERIENCE

➢ **Mechanical, Design, and R&D Engineering Intern** | Ingersoll Rand | Quincy, IL **May 2024 – Aug 2024**

- Designed and optimized essential mechanical components, utilizing additive manufacturing techniques, topology optimization, FEA, and generative design to increase product performance, longevity, and efficiency.
- Leveraged additive manufacturing techniques to design and validate novel tooling/parts for future IP and competitive advantage.
- Developed standards for Inkbit 3D printing technology and proprietary adaptive slicing software to implement industry-leading additive manufacturing techniques.
- Drove cost savings and reliability improvements by testing and identifying key design elements in competitor products.
- Completed market research, user testing, and release coordination, leading to a 7% market capture and an end-to-end launch of a new product.

➢ **President** | Helping Hand Project at NC State | Raleigh, NC **Aug 2022 – May 2026**

- Led a non-profit organization of 150+ members, providing 3D-printed prosthetics and community support to children in need.
- Managed engineering, outreach, support, and PR teams in all logistical planning and engineering practices.
- Introduced a new leadership structure and technique adoptions from CAPM/Lean Six-Sigma courses that refined processes and improved overall device quality and satisfaction by 72%.
- Coordinated meetings, events, and fundraising campaigns, boosting member engagement and resource development.
- Oversaw budgeting and financial planning to ensure sustainable operations and responsible fund management.
- Developed partnerships with local organizations and stakeholders to expand outreach and program effectiveness.

➢ **Band Logistics Manager** | NC State Department of Performing Arts and Technology | Raleigh, NC **Aug 2023 – Feb 2026**

- Managed operations and logistics for a group of over 400 marching band members in performances and events, maintaining organization and efficiency.
- Coordinated and managed the use, transport, and maintenance of equipment valued in excess of 1 million dollars.
- Built and managed a team of student staff to address member uniform, equipment, and logistical emergencies.
- Developed and maintained partnerships with NCSU Athletics to support band events and equipment operations.
- Created system to organize and consolidate student data, payments, rentals, contracts, and medical information of 400 members.

➢ **Laboratory Researcher and Assistant** | NCSU College of Veterinary Medicine | Raleigh, NC **June 2021 – May 2023**

- Conducted sample collection and processing for over 300 successful lymphoma diagnostic assays.
- Cataloged sample data and maintained electronic records, lab equipment, and inventory for a diagnostic and research lab.
- Improved and implemented an integrated billing system for sample requests.
- Created processes and protocols for processing extracted bone marrow and aspirate samples for BMT therapy research.

## RELEVANT PROJECTS

- **Senior Capstone Project: DVT-Risk Assessment Device** | Technical Lead/Project Manager **Aug 2025 – May 2026**
  - Led a multidisciplinary team through the full product development cycle for a wearable DVT risk-assessment device targeting venous occlusion risk in hospitalized patients.
  - Conducted clinical needs discovery interviews with 15+ healthcare professionals and collected survey data from 30+ clinicians to validate problem framing and use cases.
  - Initially validated a continuous glucose monitoring concept with strong clinical interest, then led a strategic pivot to DVT after identifying CGM market saturation and limited commercialization viability.
  - Performed TAM/SAM/SOM market sizing, authored DFMEA documentation, and developed design verification protocols aligned with FDA and IEC 60601 expectations.
  - Owned system architecture and implementation across modular sensor integration, electronics and PCB design, mechanical CAD, firmware for reliable signal capture, and benchtop validation of a multi-sensor leg prototype.
  - Identified that the bioimpedance sensing approach introduced additional IEC 60601 regulatory considerations, and updated design and documentation to address these requirements from the design phase onward.
  - Managed specification trade-offs and team conflict by grounding decisions in technical feasibility, manufacturability, and market constraints, while actively leveraging teammates' domain strengths (e.g., supply chain, DFM).
- **Smart Inhaler Prototype** | Personal Project **Aug 2023 – Aug 2024**
  - Designed a wearable inhaler monitoring device to address gaps in inhalation technique feedback for asthma patients.
  - Collaborated with a pulmonologist and a primary care physician to understand clinical workflow, reimbursement constraints, and adherence challenges influencing device adoption.
  - Incorporated clinical feedback to prioritize low-cost manufacturability, ease of use, and compatibility across multiple inhaler form factors, then led a small team of peers to iterate on the prototype.
- **EMG-Controlled Rehabilitation Wrist Exoskeleton** | Class Project **Nov 2025 – Dec 2025**
  - Developed servo control code and custom hardware for a forearm-mounted wrist exoskeleton enabling active and passive flexion/extension for stroke rehabilitation.
  - Performed kinematic analysis, servo calibration, and real-time EMG signal processing to achieve responsive, repeatable motion profiles during benchtop testing.
- **Wearable Microneedle Biosensor Concept** | Class Project **Nov 2025 – Dec 2025**
  - Developed servo control code and custom hardware for a forearm-mounted wrist exoskeleton enabling active and passive flexion/extension for stroke rehabilitation.
  - Performed kinematic analysis, servo calibration, and real-time EMG signal processing to achieve responsive, repeatable motion profiles during benchtop testing.
- **PL-DCS System Concept for Treatment Resistant Depression** | Class and Personal Project **April 2026 – Present**
  - Developed a concept for a PL-DCS system combining real-time EEG acquisition and processing with transcranial electrical stimulation targeting deep circuits implicated in treatment-resistant depression.
  - Researched current neuromodulation approaches for TRD and mapped where portable, closed-loop electrical stimulation could address gaps in efficacy and access.
  - Consulted clinical and research experts to evaluate clinical utility, market opportunity, and usability in both consumer and clinical environments.